



Image Captioning for Remote Sensing

Using CNN & Transformer Architectures for Remote Sensing Intelligence

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Introduction

- **Context** : The rapid expansion of high-resolution remote sensing imagery requires automated captioning systems to accelerate intelligence gathering and disaster management.
- **Domain Challenge** : Satellite images are significantly harder to process than normal photos due to their **top-down angles**, **tiny crowded objects**, and **complex spatial layouts**.
- **Technical Bottleneck** : Traditional **sequence-to-sequence** models (CNN-LSTM, CNN-GRU) process data sequentially, creating an **information bottleneck** that fails to capture the full context of an image.
- **The Proposed Architecture** : A hybrid approach where a **CNN** extracts spatial features from the image, which are then fed into a **Transformer**. The Transformer uses **Attention mechanisms** to bypass **sequential bottlenecks** and directly map those visual features to accurate text.

Objectives

- **Baseline Evaluation (Phase 1)** : Rigorously benchmark traditional recurrent models against a baseline Transformer on the Flickr8K dataset to establish architectural superiority.
- **Advance Multi-Modal Learning** : Bridge the semantic gap between computer vision and NLP by designing a custom Transformer capable of mapping complex remote sensing imagery to fluent text.
- **Architectural Optimization** : Enhance the multi-modal pipeline by integrating an EfficientNetB7 backbone, SiLU activations, increased encoder-decoder depth, multiple attention heads and Manual Label Smoothing.
- **Real-World Validation (Zero-Shot)** : Evaluate the final model's robustness through out-of-distribution (OOD) testing on unannotated, real-world satellite imagery from the 2026 Iran War.

Data Sources

Flickr8K — Generic Dataset (GitHub)

- **Role:** Generic baseline — trains all 3 candidate model architectures before domain-specific evaluation
- **Content:** 8,000 natural photographs with 5 human-written captions each
- **Model Trained:**
 - CNN-LSTM
 - CNN-GRU
 - Single-Layer Transformer
- **Purpose:** Establish baseline performance & select the best architecture for satellite fine-tuning
- **Source:** github.com/awsaf49/flickr-dataset

Satellite Image Caption Generation (Kaggle)

- **Role:** Domain-specific fine-tuning using the best model selected from Flickr8K evaluation
- **Content:** Remote sensing tiles — airports, forests, residential, industrial, agricultural zones
- **Model Trained:**
 - Single-Layer Transformer (best performer on Flickr8K)
 - Multi-Layer Transformer
- **Goal:** Produce RS-specific semantic descriptions with domain vocabulary
- **Source:** kaggle.com/datasets/tomtillo/satellite-image-caption-generation

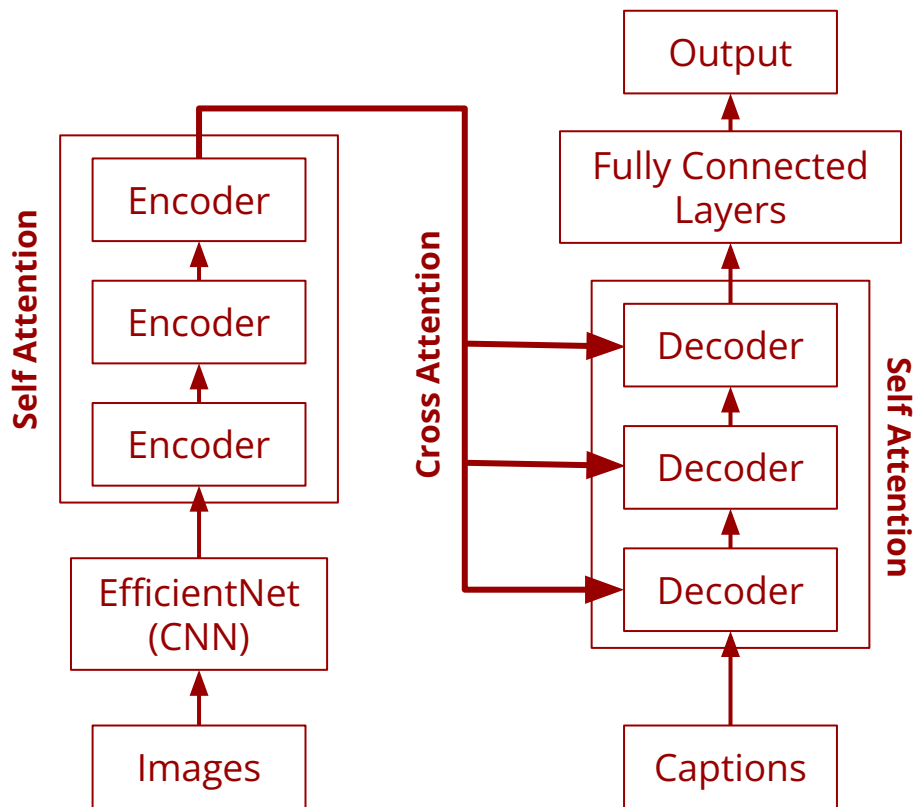
Phase 1 Results: Baseline Comparison (Flickr8K)

MODEL	VALIDATION ACCURACY	VALIDATION LOSS	GREEDY BLEU-1	GREEDY BLEU-2	BEAM (k=2) BLEU-1	BEAM (k=2) BLEU-2
CNN-LSTM	38.84 %	3.2028	55.70 %	38.93 %	60.21 %	42.64 %
CNN-GRU	40.53 %	3.1306	55.40 %	37.29 %	59.88 %	41.72 %
SINGLE LAYER TRANSFORMER	41.39 %	11.2521	58.97 %	41.35 %	60.58 %	43.12 %

Transformer Superiority:

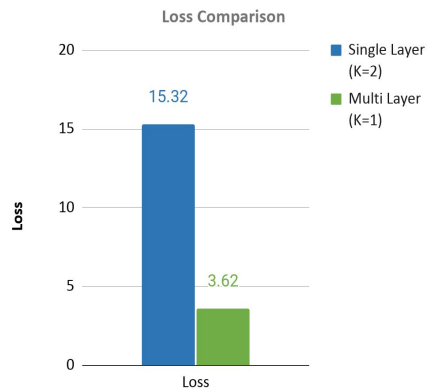
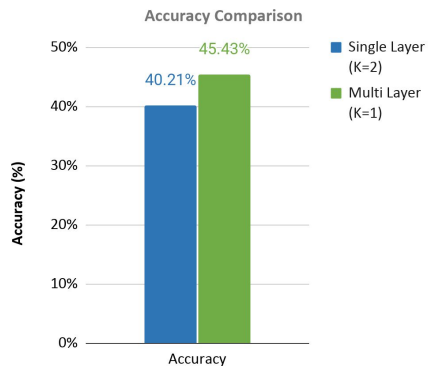
Outperformed **CNN-LSTM** and **CNN-GRU** baselines because **Self-Attention** and **Cross-Attention** directly capture long-term image-text dependencies, completely eliminating the sequential information bottlenecks inherent to **recurrent models**.

Proposed Multi-Layer Transformer Architecture



- **Vision Extractor Upgrade :** Upgraded the **CNN** backbone to **EfficientNetB7** to better capture complex spatial hierarchies. Output features are projected to a 512-embedding dimension using a Dense layer with a **SiLU (Swish)** activation.
- **Deeper Transformer Stack :** Increased architectural depth by stacking 3 **Encoder** and 3 **Decoder** blocks, utilizing 8 **Attention Heads** per block to process complex image-text dependencies.
- **Optimized Non-Linearity :** Replaced standard **ReLU** activations with **SiLU** across all **Feed-Forward Network (FFN)** sublayers to **significantly improve gradient flow**.
- **Hardware Adaptation & Regularization :** Due to the massive parameter increase, batch size was reduced from 64 to 8 to resolve **Kaggle Out-Of-Memory (OOM) errors**. Furthermore, **L2 Regularization**—which **empirically hurt performance**—was replaced with a **Dropout** rate of 0.5 to prevent overfitting.

Single Layer vs Multi Layer Transformer

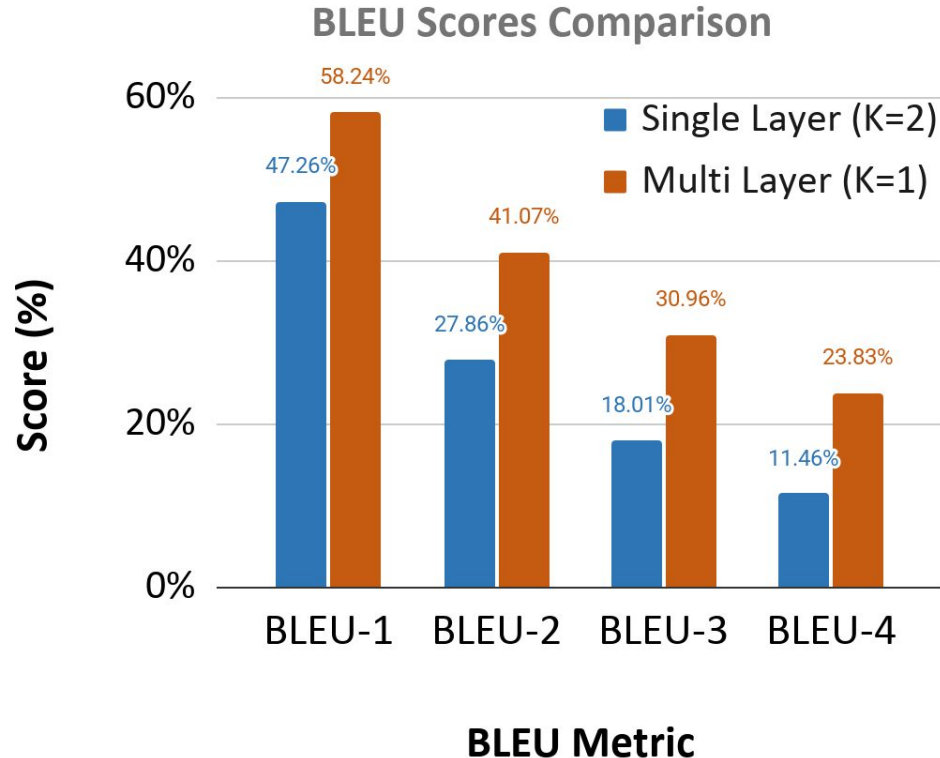


- **Significant Gains in Accuracy & Loss:** The Multi-Layer model improved accuracy by 13.0% (to 45.43%) and drastically reduced cross-entropy loss by 76.3% (from 15.31 to 3.62).
- **Universal BLEU Dominance:** The proposed Multi-Layer architecture outperformed the Single Layer baseline across all evaluation metrics and beam widths, notably achieving a **peak BLEU-4 score of 23.83%**.
- **Multi-Word Phrase Coherence:** The Multi-Layer model showed a gradual, stable score decline from BLEU-1 to BLEU-4, proving it generates coherent multi-word phrases. In contrast, the Single Layer Model's steep decline shows it struggles to maintain context beyond individual tokens.

Single Layer vs Multi Layer Transformer

MODEL	BEAM WIDTH	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Single Layer	1	48.85 %	28.06 %	17.85 %	11.37 %
Single Layer	2	47.26 %	27.86 %	18.01 %	11.46 %
Single Layer	3	45.33 %	26.38 %	16.66 %	10.21 %
Single Layer	4	44.97 %	26.35 %	16.53 %	9.94 %
Multiple Layer	1	58.24 %	41.07 %	30.96 %	23.83 %
Multiple Layer	2	57.29 %	40.50 %	30.39 %	23.23 %
Multiple Layer	3	55.63 %	39.36 %	29.68 %	22.93 %
Multiple Layer	4	53.86 %	38.05 %	28.82 %	22.39 %

Single Layer vs Multi Layer Transformer



- **Consistent Improvement** : The Multi Layer Transformer **outperformed** the Single Layer Transformer across all BLEU metrics and all beam widths without exception.
- **BLEU Scores Improve with Phrase Length** : **Improvement grows progressively** from BLEU-1 to BLEU-4, indicating more coherent and fluent multi-word caption generation.
- **Competitive with Published Baselines** : The Multi Layer Transformer's BLEU scores are **competitive** with published baselines on remote sensing image captioning tasks.
- **Collective Effect of Modifications** : The gains result from the combined effect of the **multi-layer stack**, **multiple attention heads**, **EfficientNetB7**, **SiLU** activations, **Label Smoothing**, and length-normalised **Beam Search**.

Best Performance Comparison

Metric	Single Layer (K=2)	Multi Layer (K=1)	Improvement
Accuracy	40.21 %	45.43 %	+13%
Loss	15.3162	3.6234	- 76.3%
BLEU-1	47.26 %	58.24 %	+23.2%
BLEU-2	27.86 %	41.07 %	+47.4%
BLEU-3	18.01 %	30.96 %	+71.09%
BLEU-4	11.46 %	23.83 %	+107.9%

- **Exceptional BLEU-4 Gains :** The Multi-Layer model achieved a **107.9% relative improvement** at BLEU-4, more than doubling the baseline score.
- **Significance of the BLEU-4 Metric :** As the strictest metric requiring exact 4-word sequence matches, this massive leap proves the model is not just guessing isolated words, but possesses genuine contextual comprehension and syntactic fluency.
- **Competitive Baseline Performance :** The **peak BLEU-4 score of 23.83%** is **highly competitive** with published research on satellite image captioning tasks.

Qualitative Comparison



Tokyo Olympics Stadium

**Single
Layer
Caption**

many green trees and
several buildings are
around a playground

**Multi Layer
Caption**

many buildings and
some green trees are
around a stadium



Residential Area

**Single
Layer
Caption**

some green buildings
and trees are close to a
road

**Multi Layer
Caption**

some buildings and
green trees are in a
medium residential area



Dulles International Airport

**Single
Layer
Caption**

several cars are parked
in a parking lot

**Multi Layer
Caption**

three planes are parked
on an airport

Captioning The 2026 IRAN WAR Conflict Zones



A frigate on fire at the Konarak naval base in Iran

Caption

several boats are in a port near a dock



Damage at naval facilities in Bushehr, Iran

Caption

many storage tanks are close to several buildings



A destroyed aircraft at Bushehr airport, Iran

Caption

many planes are parked in an airport near a road